MAGNETOM

Troubleshooting Guide
System

Control

Document Version

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1.1 General

Avanto systems

A description for **"Strategy"** for the AMC is deemed unnecessary since the TestTools/AMC tests are all-inclusive. When confronted with problems concerning the AMC, run the complete AMC test regardless of which component may be suspected.

The complete AMC test takes about 30 minutes.

Information about the tests can be found in the online help of the SeSo TestTools. In the event of failures, information on the result and probable cause will be given by the test programs.

Avanto Dot systems

For the dot upgrades the Tim hardware is changed the following way:

From the old AMC the following boards are removed: MPCU, PCI-TX, PCI-MON, PCI-CAN. The functionalities of these boards are taken over by the MARS and the dV_Star board.

Instead of the removed AMC boards three new boards are inserted in the rack: AMC_IO, RS232_Adapter, and CAN_Connectors. The boards supply the communication between the dV_Star and the LWL_INT together with all connections to the periphery.Besides the dV_Star board the MARS is equipped with one or more PCI_RX boards that are connected to the Receiver cassettes in the RFSU.

The following tests in TestTools are available:

Fast: Status; Control: PCI, GPUAccelaerator

Normal: Control: Performance

Interactive: PTAB Loop, CAN connector Loop, RS 232 Loop (PMU Test), FOC Walking Light

Information about the tests can be found in the online help of the SeSo TestTools. In the event of errors, information on the result and probable cause will be given by the test programs.

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1.2 General CAN Bus Troubleshooting (Avanto Systems)

In case of problems with CAN components, check the following items according to the status of the system step by step to isolate the problem.

A more detailed description of the tests may be found under the section **Procedures** . (➔ CAN Bus Procedures / Page 12).

- Check the main fuses of the components. Check LEDs on AMC/cPCI-CAN.
- Perform "Reboot Scanner".
- If unsuccessful, switch the MR system off and on again.
- Run TestTools/AMC/cPCI-CAN Comm/Status and cPCI-CAN functional.
- Run TestTools/AMC/LWL-INTComm/Status and interactive.
- Identify the affected bus: 0,1,2,or 3; check if there is activity on the bus by observing related LEDs on the optical busses (➔ - Test 2 / Page 25).
- Run TestTools/Communication tests for specific components.

  (available tests: LWL-INT, RCCS, RFIS, RFPA, PTAB, (MDSD), MSUP, SHIM, ACS, GPA).
- If communication with a specific CAN component fails, first check the following items:
  - The component is "ON" (check supply voltage LEDs)
  - Check cable connections and/or FOC connections.
  - In case of doubt, use the existing spare FOCs for the optical bus (➔ Overview CAN bus connections / Page 17).
  - Check CAN Bus error messages in the file MPCUTrace.log.
- Switch off/on the problem component; perform "Reboot scanner".
- Use TestTools/Supervision (➔ TestTools/ Supervision Mask / Page 21) to eliminate problems from suspected components.
- Refer to the specific test procedures (➔ CAN Bus Procedures / Page 12) and (➔ Interactive CAN bus test procedures / Page 16).
- Create an MR Savelog and provide it to the USC/HSC.

CAN bus error messages will be written in the file MPCUTrace.log.

This file can be found via SeSo/Reports/Process Diagnostics/MPCUTrace.log.

2.1 MPCU-5 / MPCU-6 (Diagnostic LED board)

In case of problems while booting the MPCU, you can use the Diagnostic LED board (8-LED display) to obtain additional information (error code).

> For boot problems check the battery voltage of the MPCU-5 (>2,5V) and the boot parameters, refer to “Replacement of Parts” (→ Battery of MPCU-5 / M6-020.841.13).

> MPCU-6 has the same boot parameter as MPCU-5 (→ Communication Check / M6-020.841.13), but no battery.

Required Tools

Diagnostic LED plug for MPCU-5 (part no. 4795618)

Performing the procedure

- Switch off F23 (AMC/RFSU).
- Swing out the AMC frame.
- Connect the Diagnostic LED board to X26 at the backplane, if this has not already been done.
- Switch on F23.
  - » A self test runs during boot-up of the MPCU and the steps are displayed in binary code on the 8-LED display.

If an error occurs, the error code is displayed; if the self test was successful, all LEDs are off after booting.

Example of an error code; e.g. 0b3h:

1011 0011

(the data LEDs in bold are on: **D7**-D6-**D5**-**D4**-D3-D2-**D1**-**D0**)

Give the error code to the HSC or development department for further evaluation.

In case the **MPCU hangs up in BIOS check**, the message "communication broken" is written in the savelog. Contact the HSC in this case. In case of BIOS error during boot up, a replacement of the MPCU-5 may be necessary.

AMC reset:

The AMC can be reset by pressing S1 at LWL-INT or the reset button on MPCU-5 (red LED V5 at LWL-INT lights up). To reset the AMC and RFSU, switch off/on ACC/F23.

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2.2 Exchanging PCI-TX and PCI-MON

(Avanto Systems only)

In case of problems or errors with TestTools/AMC/PCI-TX or PCI-MON, you can swap the PCI-TX and PCI-MON to get more information.

Required Tools

Normal tool kit

Performing the procedure

- Switch off F23 (AMC/RFSU)
- Exchange the PCI-TX and PCI-MON
- Switch on F23.
- Run TestTools/AMC/PCI-TX and TestTools/AMC/PCI-MON
  » You may be able to identify the defective component from the test results.

> If the external loop tests (PCI-TX/PCI-MON) fail, in particular, check the 10 MHz clock signal from the synthesizer at PCI-TX/X7.

Special **error messages** for missing 10 MHz :

MRI_PTX_PTX_GC_FIFO_OVL

MRI_GCD_PCIGC_CLOCK10_ERROR

The last error message displays if you want to start a measurement and the 10 MHz signal is missing at PCI-TX.

Conclusion

If the 10 Mhz signal is ok, the PCI-TX is probably defective. If not, check the synthesizer and the cabling between RFSU/synthesizer/X24 (CLK10_M) and PCI-TX/X7.

2.3 Control/ PCI

(Avanto Dot Systems only)

2.3.1 General

The **PCI** test executes the external component tests of all peripheral hardware components supporting self-tests. The available links are used for this. Therefore, this test checks the performance and the optical power of the communication links.

For the DOT upgrades, the Tim hardware is changed as follows:

From the old AMC, the following boards are removed: MPCU, PCI-TX, PCI-MON, PCI-CAN.The functionalities of these boards are taken over by MARS and the dV_Star board.

In place of the removed AMC boards, three new boards are inserted in the rack: AMC_IO,RS232_Adapter, and CAN_Connectors. The boards supply the communication between dV_Star and LWL_INT together with all connections to the periphery.

Besides the dV_Star board, MARS is equipped with one or more PCI_RX boards that is (are) connected to the receiver cassettes in the RFSU.

Before starting the real test, it is first checked whether MARS is reachable via network, ("ping"). If this fails, a message is printed out and the program terminates immediately; the other component is also not reachable.

For each executed test, the test results will be shown, including the name of the test, its status, and a message showing the test details.

The test execution will be continued even if an error occurs when executing a test.

2.3.2 Tests

2.3.2.1 AMC_IO

TestTools/ Fast/ Control/ PCI/ AMC_IO

- **DACCableTest**

  The cabling including the input at the gradient amplifier control (GSSU) is tested for a shortcut.

> It is not recognized if the cable is not plugged in!

DACIfTest

The cabling including the GSSU is tested for a shortcut.

> Here it is recognized that the cable is not plugged in!

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ADCIfTestDataLines

A command is sent to the GSSU and a pattern is read back, but no pattern is written into the GSSU.

ADCIfTestCtrlLines

A command is sent to the GSSU, a pattern is written and read back, band finally evaluated.

- **LightIntensity**

  The transmitted and received light intensity/power of the optical transceivers are tested and evaluated against a specification value/threshold.

- **SupplyVoltage**

  The supply voltages of the optical transceivers are checked against specification.

- **Temperature**

  The temperatures of the optical transceivers are checked against specification.

- **AMCLoop**

  The LED registers are written and their status read back (pattern).

2.3.2.2 Signal Path

Communication between MARS and the peripheral components is performed using the AMC_IO and the dV Star.

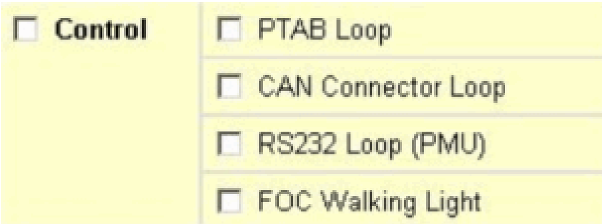
2.4 TestTools/ Interactive/ Control

(Avanto Dot Systems only)

2.4.1 Tests

TestTools/ Interactive/ Control

Fig. 1: Interactive Tests AMC



2.4.1.1 PTAB Loop Test

The test simulates the position signals coming from the MDSD for patient table movements while a sequence is running. It is checked whether the counters are counting up and down correctly.

For the test, two plastic fibers of the test cable set are necessary. They are used as loop cables to create a kind of "shortcut".

The following connections have to be established in the electronics cabinet at the AMC and the LWL_INT D6 (board) located there.

- Fiber Optic Converters U4 with U19 and

- Fiber Optic Converters U5 with U20.

After pressing OK, the test will start.

Signal Path Only the AMC is involved.

2.4.1.2 CAN Connectors Loop Test

The test simulates the CAN communication between CAN0 and CAN1 and between CAN2and CAN3.

For the test, two SUB-D 9 pin test cables of the test cable set are necessary. They are used as loop cables to create a kind of "shortcut".

With these, the following connections have to be made in the electronics cabinet in the AMC at CAN_Connectors D21:

- CAN0 with CAN1 and

- CAN2 with CAN3. After pressing OK, the test will start.

Signal Path

dV-Star - Sig_distr2 - CAN_Connectors - (backplane - LWL-Int).

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2.4.1.3 RS 232 Loop (PMU) Test

For the test, onr plastic fiber of the test cable set is necessary. It is used as loop cable to create a kind of "shortcut".

With this the following connection has to be made in the electronics cabinet in the AMC at LWL_INT D6:

- Connect at LWL_INT U6 with U7 using a fiber optic cable.

Additionally, the physical display and the PMU beeper at the Intercom have to be switched on.

After pressing OK, the test will start.

Signal Path

dV-Star - Sig_distr2 - RS232-Adapter - backplane - LWL-Int.

2.4.1.4 FOC Walking Light Test

The test outputs a bit pattern to the TX LEDs (fiber optic cable/FOC transceivers) of the LWL_INT board (U1 to U3 and U14 to U27), and of the AMC_IO (U1 through U5, and U7).

First you have to unplug these FOCs, do not unplug U6 to U13 of the LWL_INT (PMU and CAN bus).The test runs as follows:

Step 1: all LEDS of the FOC transceivers will go out;

Step 2: beginning with U1 of LWL_INT, ending with U7 of the AMC_IO, the LEDs will light on and off one after the other;

Step 3: all LEDs light up;

Step 4: all LEDs return to the initial status.

After pressing OK, you will have approx. 60 seconds to go to the ACC cabinet and observe the FOC transceivers.

Signal Path Only the AMC is involved.

At the end, the result of the test has be confirmed with a pop-up. For this interactive test, automatic evaluation is not possible.

After the test has been finished, the scanner has to be rebooted unless the test is repeated immediately.This can be done manually or after confirming the corresponding pop-up.

2.5 CAN Bus Procedures

(Avanto Systems only)

2.5.1 Procedures using TestTools/ AMC

2.5.1.1 cPCI-CAN (Comm/Status)

- Run TestTools/AMC/cPCI-CAN
  - » The communication to the following CAN components is tested and the component status displayed:

CAN bus 0 (open, mute, electrical/optical):

cPCI_CAN <-> X120/Alarm Box & FOC Converter/ X126

<-> MSUP <-> TALES (TaTS only) <-> RCCS <-> RFIS <-> PTAB

CAN bus 1 (open, active, electrical):

cPCI-CAN - LWL-INT - SHIM - ACS - GPA

CAN bus 1 (open, active, optical):

MDSD (i.e. Move During Scan Drive in patient table)

CAN bus 2 (optical):

cPCI-CAN - LWL-INT - RFPA1/2

CAN bus 3 (electrical/SaTS only):

cPCI-CAN - GPA

Conclusion

If the communication test of a component fails, the power supply of the component, the cable/FOC connection to the LWL-INT, the component/SLIO, the LWL-INT or the cPCI-CAN may be defective.

Also run the LWL-INT (LED) interactive test.

2.5.1.2 cPCI-CAN (Functional)

- Run TestTools/AMC/cPCI-CAN
  - » The test checks whether at least one component from a particular bus (0, 1, 2 or 3) responds.

Conclusion

If no component responds, the LWL-INT is probably defective. Run the LWL-INT interactive test.

If the RFPA1/2, (RFPA 2 only with spectroscopy option installed) on CAN bus 2 does not respond, check the RFPA1/2 and its FOC connection lines; the RFPA1/2 is the only component on CAN bus 2.

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If a component of the optical bus does not respond, check whether the adjacent component in direction of the FOC-converter X126 is ok.

2.5.1.3 TestTools/ Fast/ Control/ PCI AMC_IO K2268 (VD13A only)

- **DACCableTest**

  The cabling including the input at the gradient amplifier control (GSSU) is tested for a shortcut.

> It is not recongized if the cable is not plugged in!

DACIfTest

The cabling including the GSSU is tested for a shortcut.

> Here it is recognized that the cable is not plugged in!

ADCIfTestDataLines

A command is sent to the GSSU and a pattern is read back, but no pattern is written into the GSSU.

ADCIfTestCtrlLines

A command is sent to the GSSU, a pattern is written and read back, band finally evaluated.

- **LightIntensity**

  The transmitted and received light intensity/power of the optical transceivers are tested and evaluated against a specification value/threshold.
- **SupplyVoltage**

  The supply voltages of the optical transceivers are checked against specification.
- **Temperature**

  The temperatures of the optical transceivers are checked against specification.
- **AMCLoop**

  The LED registers are written and their status read back (pattern).

2.5.1.4 LWL-INT (Interactive LED Test)

Procedure

- Run TestTools/AMC/LWL-INT; follow the instructions from the program:

  First remove all FOCs, except U6 to U13 (PMU and CAN bus).

  » The test outputs a bit pattern to the TX LEDs (U1 to U3 and U14 to U27) on the LWL interface (running LED light):

  The test runs as follows:

1. The test outputs a bit pattern to the TX LEDs (U1 to U3 and U14 to U27) on the LWL interface (running LED light).
2. All LEDs U1 to U3 and U14 to U27 will be switched off.
3. Beginning with U1, the LEDs will be switched on and off one after the other.
4. All LEDs light up.
5. All LEDs will be switched off.

Conclusion

If one of the LEDs does not light up or the LEDs are significantly dimmed, replace the LWL-INT board.

Also check the PCI-TX board.

Tab. 1 FOC connections at LWL-INT

FOC connection	Function
U1: RFPA1_UNB	Light on: RFPA1 unblanked; signal comes from PCI-TX
U2: RFPA2_UNB	Light on: RFPA2 unblanked; signal comes from PCI-TX
U3: FAST_EXT_TRIG_OUT	Light on: Trigger
U4: INC A/MDS_PTAB	Increment sender signals PCI-TX
U5: INC B/MDS_PTAB	Increment sender signals PCI-TX
U6: PMU_TX	Serial interface to PMU
U7: PMU_RX	Serial Interface to PMU
U8: CAN0_TX U9: CAN0_RX	CAN bus 0 to FOC (optical bus); **can be used for troubleshooting optical CAN bus 0.**
U10: CAN1_TX U11: CAN1_RX	CAN bus 1 via FOC to MDSD U5/RX, U6/TX
U12: CAN2_TX U13: CAN2_RX	CAN bus 2 via FOC to RFPA U2/TX, U3/RX
U14: RFIS_SEL3 U15: RFIS_SEL2 U16: RFIS_SEL1 U17: RFIS_SEL0	Selection of registers on RFIS; signal is generated on PCI-TX on a pulse start command
U18: N_SEQ_ON(RFIS)	Light off: Sequence running; signal from PCI-TX
U19: DYN_CTRL	Switching between tune and detune; signal is generated on PCI-TX on a pulse start command.
U20: LC_PICKUP_SEL	Light on: Local coil selected; signal from PCI-TX

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FOC connection	Function
U21: SE_CTRL	Switching of TX/RX-switch; signal is generated on PCI-TX on a pulse start command.
U22: RCCS_SEL3	Selection of registers on RCCS; signal is generated on PCI-TX on a pulse start command.
U23: RCCS_SEL2	
U24: RCCS_SEL1	
U25: RCCS_SEL0	
U26: LOW_GAIN	Switching of preamplifiers of RCCS to low gain; signal from PCI-TX
U27: SEQ_ON (User)	Light on: Sequence is running
U28: RESERVE_LWL_IN1	Reserve

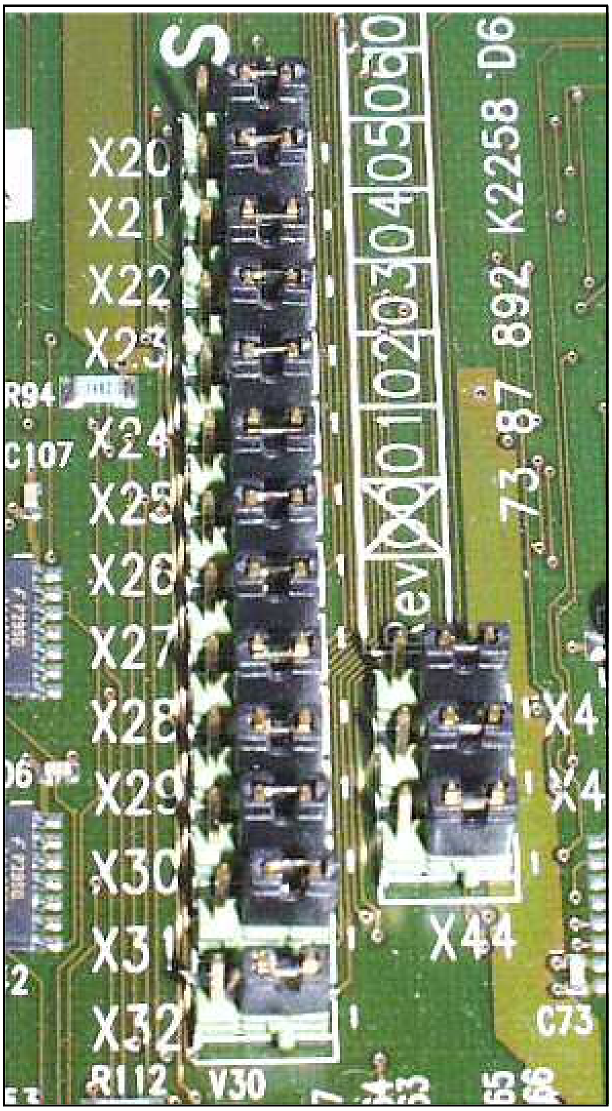
Voltage control LEDs on LWL-INT

Tab. 2 LEDs on LWL-INT

LED	ON
V1 green	+ 3.3 VDC ok
V2 green	+ 5 VDC ok
V3 green	+12 VDC ok
V4 red	-12 VDC not ok
V5 red	- during AMC Reset pressed - during power-up reset - in case of voltage error (+3.3 V, +5 V, +12 V)

Check of jumpers on LWL-INT

Fig. 2: Jumpers on LWL-INT



Jumpers X20 - X32 and X42 - X44 in position 1 - 2.

2.5.2 Interactive CAN bus test procedures

2.5.2.1 General

Electrical CAN bus:

All components on the bus are connected in parallel on the bus. If one component has no power, all other components are active on the bus and will respond.

Therefore, a specific suspicious component can be switched off or excluded from the bus by connecting the in/output connectors together. The supervision of the component can be changed using TestTools/Supervision.

Optical CAN bus:

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All components on the bus are connected in series. If a component has no power or its CAN interface fails, all components behind this unit cannot communicate with the controller via the bus.

The suspicious component can be excluded from the bus by connecting the FOCs of the unit in front of it (as seen from the AMC) with the component behind it (as seen from the AMC).

TX is always connected with RX (and vice versa) of two adjacent units.

2.5.2.2 Overview CAN bus connections

The following tables show the CAN bus connections of all CAN units of the specific busses.

Additionally, the following spare FOCs from the ACC to the magnet are available:

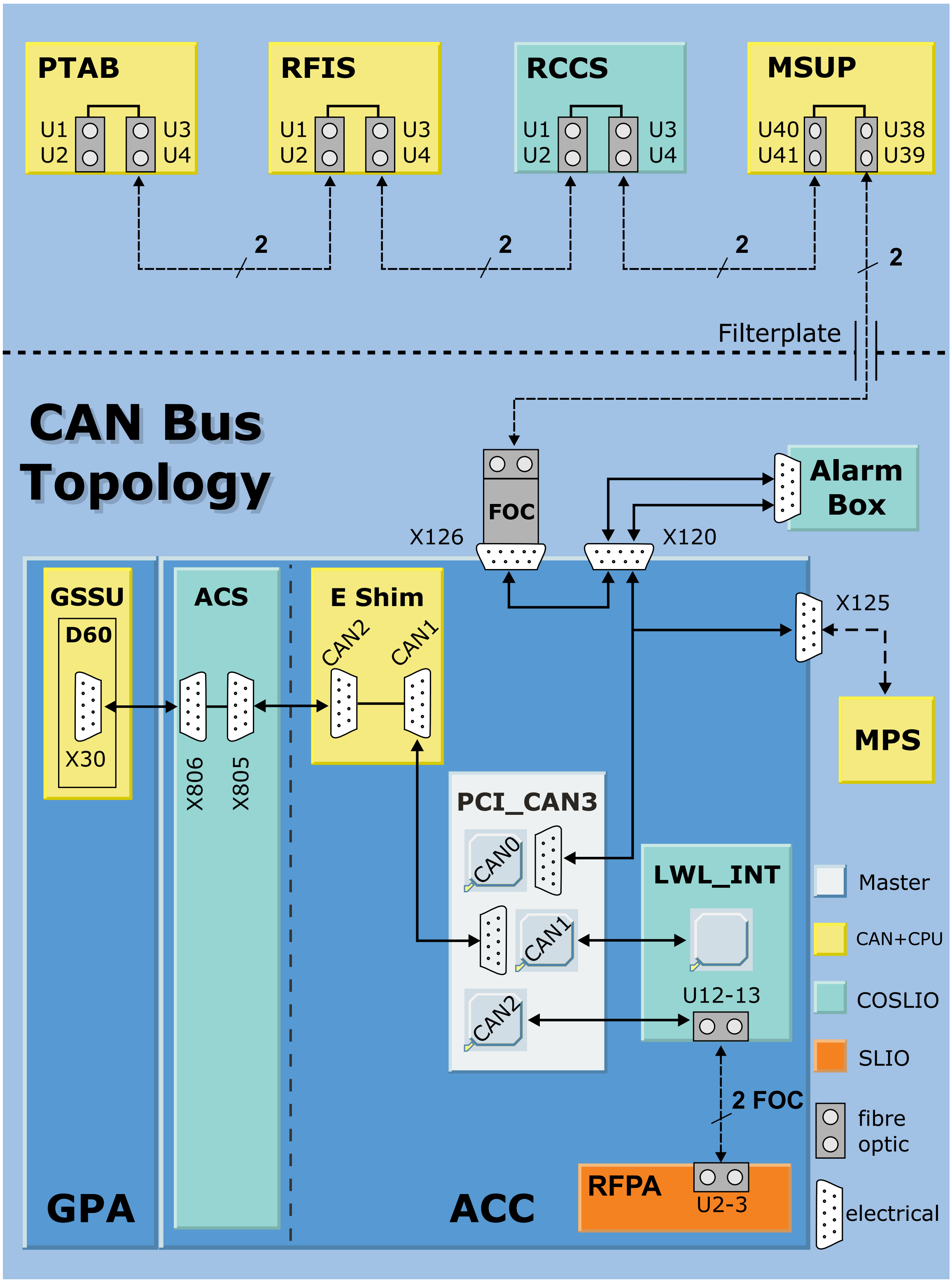
W1420 (AMC/PCI-TX <-> TALES): 1 spare FOC

W1421 (AMC/LWL-INT <-> RFIS): 1 spare FOC

W1425 (AMC/LWL-INT <-> RCCS): 1 spare FOC

W 1426 (AMC <-> Astex 35 kW RFPA); 2 spare FOCs (Trio only)

Fig. 3: CAN Bus Topology for Avanto / Espree



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Tab. 3 Avanto/Espree/Trio (CAN 1)

AMC - cPCI CAN 1 (CAN open, active, electrical)				
cPCI CAN 1 electrical bus <->				LWL-INT optical bus U10 TX -> U11 RX <-
LWL-INT (CO-SLIO)	SHIM (MCB3) CAN2 <-> CAN1 <->	ACS (KKW) X 805 <-> X 806 <->	GPA (MCB3) <-> X30 Term.res.	MDSD (JAT) U5 RX <- U6 TX ->

Tab. 4 Avanto/Espree/Trio (CAN 0)

AMC - cPCI CAN 0 (CAN open, mute, electrical/optical)				
cPCI CAN 0 electrical bus <->				LWL-INT optical bus U8 TX -> U9 RX <-
LWL-INT (CO-SLIO)	MPS (MCB3) (only for magnet ramping)	to X120 from X120/ ACC to Alarm Box (CO-SLIO) from X120 to FOC converter X126 (FOC converter: TX right, as seen from the front of ACC)		Espree only: RFPA (CO-SLIO) U2 TX -> U3 RX <- (CAN1)
Optical bus from FOC converter X126 or from LWL-INT optical bus TX U8 -> and RX U9 <-				
MSUP (MCB3) from/to X126: U39 RX <- U38 TX -> from/to TALES: U41 RX <- U40 TX ->	TALES (CO-SLIO) from/to MSUP: U6 RX <- U7 TX -> from/to RCCS: U4 RX <- U5 TX ->	RCCS (CO-SLIO) from/to TALES: U4 RX <- U3 TX -> (CAN2) from/to RFIS: U2 RX <- U1 TX -> (CAN1)	RFIS (2xCOSLIO) from/to RCCS: U3 RX <- U4 TX -> (CAN2) from/to PTAB: U1 RX <- U2 TX -> (CAN1)	PTAB (JAT) from/to RCCS: U4 RX <- U3 TX ->

Tab. 5 Avanto/Espree (CAN 2)

AMC - cPCI CAN 2 (CAN prop, mute, electrical/optical)	
cPCI CAN 2 electrical bus	LWL-INT optical bus from/to RFPA: U12 TX -> U13 RX <-
LWL_INT (CO_SLIO)	RFPA (SLIO) from/to LWL-INT U2 TX -> U3 RX <-

Tab. 6 Symphony TIM (CAN 3)

AMC - cPCI CAN 3 (CAN prop, active, electrical/ optical)	
cPCI CAN 3 electrical bus	GPA D139 (CAN piggy) optical bus U3 TX -> U4 RX <-
-> GPA/ D16 X401 GPA/ D16 X402 -> GPA/ D139 (CAN piggy)	RCA (CAN piggy) U14 TX -> U15 RX <-

Tab. 7 Symphony (CAN 1)

AMC - cPCI CAN 1 (CAN open, active, electrical/optical)		
cPCI CAN 1 electrical bus		LWL-INT optical bus U10 TX -> U11 RX <-
LWL_INT (CO-SLIO)	SHIM (MCB3)	MDSD (JAT) U5 RX <- U6 TX ->

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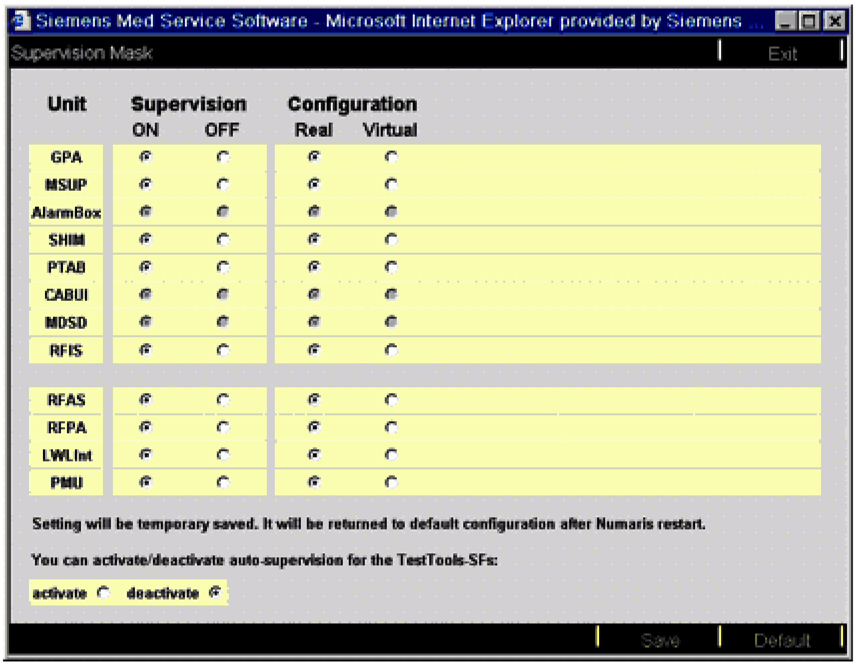
Tab. 8 Trio TIM (CAN 2)

AMC - cPCI CAN 2 (CAN prop, mute, electrical/optical)		
cPCI CAN 2 electrical bus	LWL-INT optical bus from/to RFPA 35 kW U12 TX -> U13 RX <-	
LWL-INT (CO-SLIO)	RFPA 35 kW (SLIO) optical bus from/to LWL-INT: U2 TX -> U3 RX <- (CAN1) optical bus from/to RFPA 8 kW: (with spectroscopy option only) U4 TX -> U5 RX <- (CAN2)	RFPA 8 kW (SLIO) optical bus from/to RFPA 35 kW U3 RX <- U2 TX -> (with spectroscopy option only)

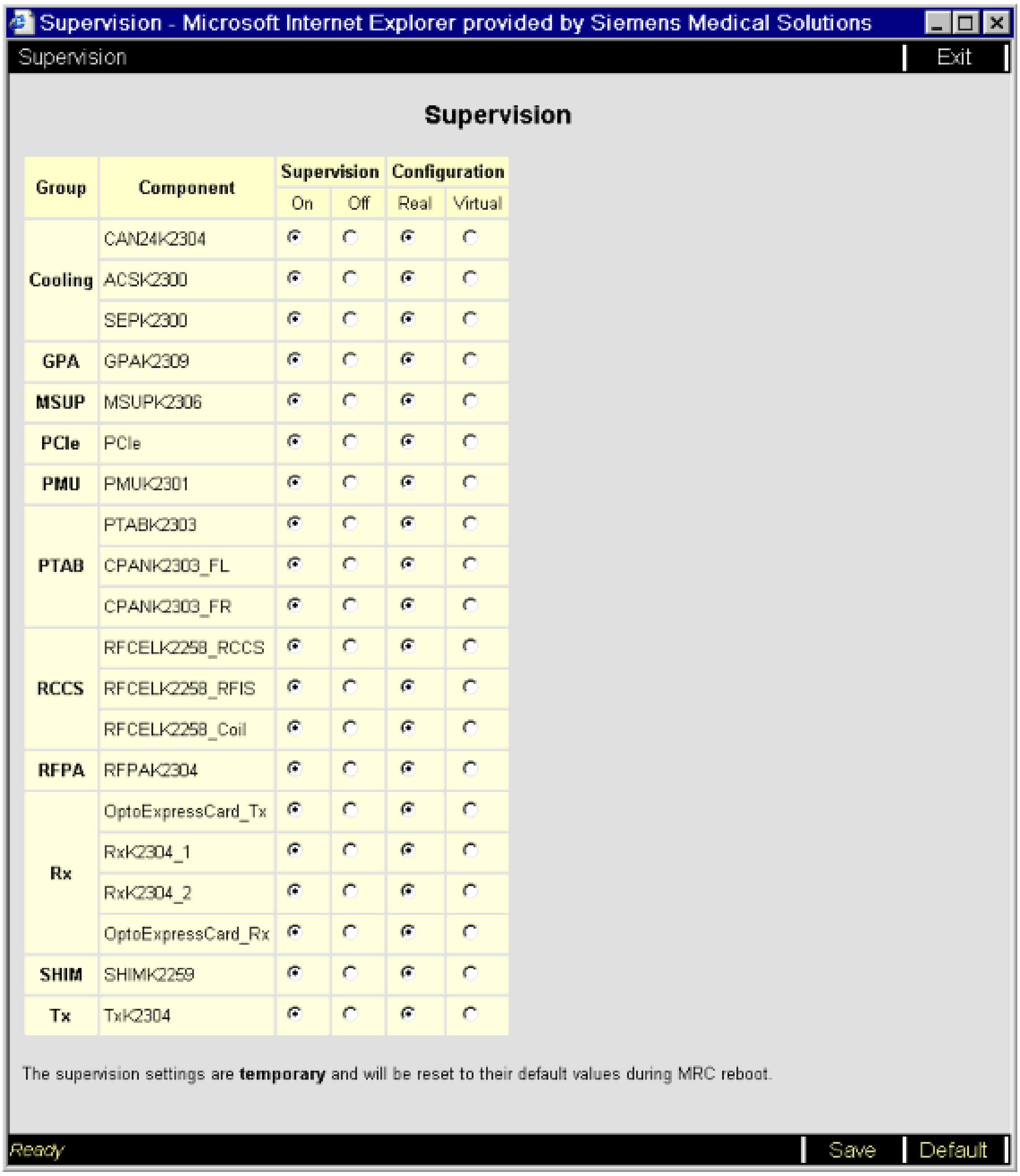
2.5.2.3 TestTools/ Supervision Mask

Fig. 4: Supervision Mask



TestTools/ Supervision Mask in software version VB13A.

Fig. 5: Example TestTools "Supervision" window (VD13A)



Supervision - Microsoft Internet Explorer provided by Siemens Medical Solutions

Supervision | Exit

Supervision

Group	Component	Supervision On	Supervision Off	Configuration Real	Configuration Virtual
Cooling	CAN24K2304	◉	○	◉	○
	ACSK2300	◉	○	◉	○
	SEPK2300	◉	○	◉	○
GPA	GPAK2309	◉	○	◉	○
MSUP	MSUPK2306	◉	○	◉	○
PCIe	PCIe	◉	○	◉	○
PMU	PMUK2301	◉	○	◉	○
PTAB	PTABK2303	◉	○	◉	○
	CPANK2303_FL	◉	○	◉	○
	CPANK2303_FR	◉	○	◉	○
RCCS	RFCELK2258_RCCS	◉	○	◉	○
	RFCELK2258_RFIS	◉	○	◉	○
	RFCELK2258_Coil	◉	○	◉	○
RFPA	RFPAK2304	◉	○	◉	○
Rx	OptoExpressCard_Tx	◉	○	◉	○
	RxK2304_1	◉	○	◉	○
	RxK2304_2	◉	○	◉	○
	OptoExpressCard_Rx	◉	○	◉	○
SHIM	SHIMK2259	◉	○	◉	○
Tx	TxK2304	◉	○	◉	○

The supervision settings are **temporary** and will be reset to their default values during MRC reboot.

Ready | Save | Default

The supervision status should be changed whenever individual components are not to be monitored during troubleshooting.

- ON:

The component is supervised and the error messages from the peripheral units are monitored (normal setting).

- OFF:

The error messages from the peripheral units are not monitored, but the data transfer and the acknowledge from the component are still supervised. Use this mode to suppress incoming error messages from the peripheral units.

- Real:

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The component is installed and supervised (normal setting).

- Virtual:

The component is treated as existent, but the data transfer to and the acknowledge from the component is only simulated. This mode should be used for troubleshooting when a test measurement must be started without supervision of a certain component.

Clicking the **Default** button displays the default configuration. Click **Save** to activate the settings temporarily. **The default configuration will be restored after syngo MR is re-started.**

2.5.2.4 Procedures for electrical CAN bus

- Test 1

- **Excluding suspected components from the CAN bus**

Procedure

- Using TestTools/ SupervisionMask (Ref), switch the supervision of the suspected component to off and configuration to virtual.
- Check the communication to every other component on the specific bus using TestTools, Comm/Status.
- Switch off the suspected component and check the communication to the other components using TestTools.
- If the communication to the other components of the specific bus is ok, perform Test 2.

- Test 2

- **Test of suspected component**

Procedure 1

- Using TestTools/ Supervision Mask (Ref), switch the supervision of all but the suspected component to off and configuration to virtual.
- Switch off all but the suspected component.
- Check the communication to the suspected component on the specific bus using TestTools, Comm/Status.

Procedure 2

- Connect only the suspected component to the bus (switch all other components off and/or disconnect all bus cables from all other components and connect the input and output connection together).
- Perform "Reboot scanner".
- Run TestTools, Comm/Status for the suspected component.

- Test 3

- **Special cases**

- Error messages from ACS controller

Procedure

- Check the hardware version of the ACS controller.
- Check function of Chiller/SEP/IVP.
- Check connections between Chiller/SEP/IVP and ACS.
- Replace ACS controller.

- **Interference between ACS and MDSD controller**

Sometimes the error messages of many components affect the bus so that is no longer possible to detect the component causing the original error (Example: interferences between the ACS controller on electrical bus 1 and the MDSD controller on optical bus 1).

Procedure

- Using TestTools/ Supervision Mask (Ref), switch the supervision of PTAB/ CABUI/ MDSD to off and configuration to virtual.
- Run TestTools/ ACS, Comm/Status.

  If errors from ACS are displayed, continue with Test 3 "Error messages from ACS controller"
- Switch Supervision Mask of PTAB/ CABUI/ MDSD back to on and real.
- Check FOCs from LWL-INT CAN 1 (U10/TX, U11/RX) to table electronics (remove the front cover of the table electronics), MDSD (U5/RX, U6/ TX). In case of doubt, use two spare FOCs.
- Switch off ACS and perform "Reboot scanner".
- If MDSD is not ready again or fails during TestTools/MDSD, Comm/Status Test, replace the MDSD controller.

- Check bus termination on GPA (Symphony TIM)

- Check the bus termination jumper on GPA/ GSSU/ D 139 X3 left:

  Position 1-2/ terminated.

2.5.2.5 Procedures for optical CAN bus

> Do not look directly into the bright transmitter LEDs; watch them sideways to avoid their effect.

- Test 1

- **Check FOC converter (MR system off)**

Procedure

The MR system is switched off. The MSUP and alarm box are permanently connected to the mains. The FOC converter X126 on the top of the ACC is supplied via the MSUP/Alarm Box. The FOC converter is connected via FOCs to the MSUP (➔ Tab. 4 Page 19).

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- Remove the right TX/FOC (as seen from the front of the ACC) from X126.

  » The red transmitter LED of the FOC converter must be on.

  Reconnect the FOC.
- Remove the left RX/FOC (as seen from the front of the ACC) from X126.

  » The RX/FOC must be showing red light.
- Leave the RX/FOC disconnected for more than 10 seconds. The alarm box watchdog triggers the "COMMS FAULT". Reconnect the RX/FOC and reset the alarm on the alarm box.

  Possible causes of (sporadic/permanent) errors:

  - Voltage supply for X126 is too low
  - Defective FOCs between X126 and MSUP
  - Defective X126; poor mechanical installation on top of ACC (also causes error messages from components in the following bus chain).

- Test 2

- **Check CAN bus activity at components**

Procedure

- Repeatedly run TestTools, Comm/Status for the affected CAN bus/ component.
- Check TX/RX LEDs activity for specific components:

Tab. 9 CAN bus control LEDs

Component	Control LED
TALES	DATA IN/ DATA OUT CTOG
RCCS	CAN2 / CAN1
RFIS	CAN_TX1 CAN_TX2 CAN_RX1 CAN_RX2

- Test 3

- **Connection of optical Can bus components directly to LWL-INT**

  Remember to always connect RX to TX (and vice versa) for two adjacent components.

Procedure 1

In case of problems on the optical bus (CAN 0), e.g. disturbances from MSUP, problems with the FOC converter (X126 on top of the ACC), or supposed defects of FOCs between X126 and MSUP, connect the components after the MSUP directly to LWL-INT (U8/TX, U9/RX) using the spare FOCs:

- Disconnect FOCs U4 and U3 at RCCS (from MSUP) and connect LWL-INT U8, U9 to RCCS U3, U4 using the spare FOCs.

- Perform "Reboot scanner".
- Run TestTools/"component", Comm/Status.

Alternatively:

- Connect the spare FOCs from LWL-INT to the first component on the optical bus that does not respond.
- Perform "Reboot scanner". Check which components respond.
- Run TestTools/"component", Comm/Status.

Procedure 2

To identify the defective component, proceed as follows:

- Disconnect the FOCs of the suspected component.
- Connect the FOCs of the two neighboring components.
- Connect the spare FOCs from LWL-INT to the suspected component.
- Perform "Reboot scanner". Check which components respond.
- Run TestTools/"component", Comm/Status.

- Test 4

- **Test procedure for sporadic and/or permanent errors of CAN bus FOC receivers (RFIS, RCCS, TALES)**

Procedure

Preparations:

- Position the large spherical phantom with loader in the magnet center.
- Preparation of measurement:

  - Start the Service Software UI (Options/service/local service)

  - Start TestTools/Tools

  Click radio buttons **Service Patient Registered** and **Adjustment Disabled**

  - Leave the platform with **save**.
- Load the service sequence rf_pulse from ExamExplorer into the protocol control region:

  `-> Sequence Region -> Service Sequences -> Default Protocols`
- After the sequence is "Open", you can change parameters in the parameter card:
- Change the following parameters in the parameter cards:

  Sequence Special:

  - RF Pulse Length: 5 ms

  - Loop Mode: TX -> PFPR1 -> TAS

  System/ Transmitter/ Receiver:

  -sSRF01 [1H]: 0 V (Check and adjust this value before every measurement series)

  Contrast:

  - Measurements: 300

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- Open the door of the ACC and search for a spare FOC: <W1420>, <W1421> or <W1425> (left of AMC).
- Start the sequence.
- Pull the FOC from U1 AMC/LWL-INT (D6). You should see the red flashing output LED (Signal RFPA_UNBLANK).
- Find the corresponding end of the spare FOC (from ACC to magnet) to test the FOC receivers of TALES, RCCS and RFIS:

  At the magnet side, one of them ends up on top of RFIS, the other somewhere left to the TALES/BCCS. It may be necessary to cut some cable ties to get the appropriate length to reach the FOC inputs to be tested.
- If not labeled, connect the spare FOCs one after the other to D6/U1(AMC) and check the other side at the magnet for flashing output. Mark the corresponding ends of FOCs for later use.

Test of FOC receivers:

> If the sequence stops during testing, the system has to be switched off/on (system shut down/system off/system on).

- Remove the FOCs from the inputs of a single component shown in the following tables (top of TALES and RFIS, left side of RCCS).
- Connect the flashing end of the spare FOC one after another to every input. The corresponding Control LEDs should be flashing according to the input (refer to the tables below).
- Hold the fully inserted FOC connector at the back end - not at the FOC - and with a circular movement of the hand gently stress the connection mechanically.

  » If the Control LEDs do not flash but are either permanently off or on during the manual movement, the component has to be replaced.
- Reconnect all FOCs at the component.

Input	Control LED
U6 RX CAN2 (from MSUP)	DATA IN/ DATA OUT
U4 RX CAN1 (from RCCS)	DATA IN/ DATA OUT

Tab. 10 Input connections and control LEDs of RCCS

Input	Controll LED (yellow)
U2 RX CAN1 (from RFIS)	CAN2
U4 RX CAN2 (from MSUP)	CAN1

Tab. 11 Input connections and control LEDs of RFIS

Input	Controll LED (yellow)
U1 RX CAN1 (from PTAB)	CAN TX2/RX1 and weaker TX1
U3 RX CAN2 (from RCCS)	CAN TX1/RX2 and weaker TX2

Final work:

- Reconnect the original FOC at AMC/LWL-INT (D6), (RFPA_UNBLANK).
- Stop the sequence (STOP button).
- Attach the protecting caps to the spare FOCs and secure them with cable ties.
- Switch the MR system off and on again (system shut down/system off/system on).
- Run a normal image measurement (spherical phantom with loader).

- Test 5

- **Remove suspect components causing interferences from the CAN bus.**

Procedure

- Beginning from PTAB, disconnect one after the other the following components (by disconnecting the FOCs to the following component) until the normal function of the bus is established.
- Run TestTools/"component", Comm/Status.
- Also perform Test 3.

2.5.2.6 Additional Information

- Using the program "percom"

Procedure

WARNING

When using "percom", it is possible to set the MR system to a mode where patient safety modules are effectively disabled.

Measuring with these safety modules disabled may lead to permanent injuries. It is also possible to measure using an incorrect image position.

» **Only specially trained persons with knowledge of these risks are permitted to use "percom".**

With "percom", you have direct control of specific components.

As an example, you can get information about firmware and application software of CAN components with the command **"get HW info"** in the sub-menu of the specific component, checking and setting of parameters, etc.

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You need a password to use "percom": Call the HSC/MR group in Erlangen to create the password:

2.5.2.7 - Creating the Password

- When starting the tool (typing “percom” in a command prompt) for the first time, type in a ’?’ when you are prompted for the password.
- A key such as ’as95np/o’ will be returned.
- Copy the key and send it via email to the HSC/MR group in Erlangen to create the password (the password has a limited period of validity for the system for which the key was generated).
- After you get the password, restart the tool and type in the specific password.

2.5.2.8 - Replacement of CAN interface/Component

The table below shows the replaceable components of the CAN bus. Some system components have to be replaced completely.

(For the replacement of individual components refer to the manual "Replacement of parts").

Tab. 12 Replaceable CAN bus components

CAN - component	System component (connected to CAN bus)
AMC: cPCI-CAN AMC: LWL-INT ACC: FOC converter ACS: Controller GPA/GSSU: D60 / D139 (Symphony TIM only) MKS RFPA: CAN bus controller	Shim unit RFPA (DORA/Avanto/Symphony TIM, PORA/ Espree, Dressler/ Trio TIM) Alarm box
PTAB: MDSD device A 4320 RCA: Controller RFIS: CAN_AND_CODE_CTRL PTAB: Controller A 4140	MSUP TALES RCCS

2.5.2.9 - Service Tools available for FOCs

To repair broken FOCs and/or connectors:

LWL REP-SET 83 97 569

For measuring the damping of a FOC:

Fibre Optic Testing Device 88 99 200

2.6 CAN Bus Procedures

(Avanto Dot systems only)

2.6.1 CAN Connectors Loop Test

TestTools/ Interactive/ Control/ CAN Connector Loop

2.6.1.1 General

The test simulates the CAN communication between CAN0 and CAN1 and between CAN2 and CAN3.

For the test two SUB-D 9 pin test cables of the test cable set are necessary. They are used as loop cables to create a kind of "shortcut".With these the following connections have to be made in the electronics cabinet in the AMC at CAN_Connectors D21:

- CAN0 with CAN1 and

- CAN2 with CAN3

2.6.1.2 Signal path

dV-Star - Sig_distr2 - CAN_Connectors - (backplane - LWL-Int)

2.6.1.3 Conclusions

Refer to the diagnosis of the test within TestTools.

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Fibre Optic Testing Device 88 99 200 Partnumber changed

Version 03

- MPCU-5 Battery voltage check note added.

Version 04

- PCI AMC_IO tests for Avanto Dot systems added.
- CAN Connectors Loop Test for Avanto Dot systems added)
- Minor changes to support D13.

Version 05

- MPCU-6 added (➔ MPCU-5 / MPCU-6 (Diagnostic LED board) / Page 6)

4 List of Hazard IDs

There are no Hazard IDs in this document.

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